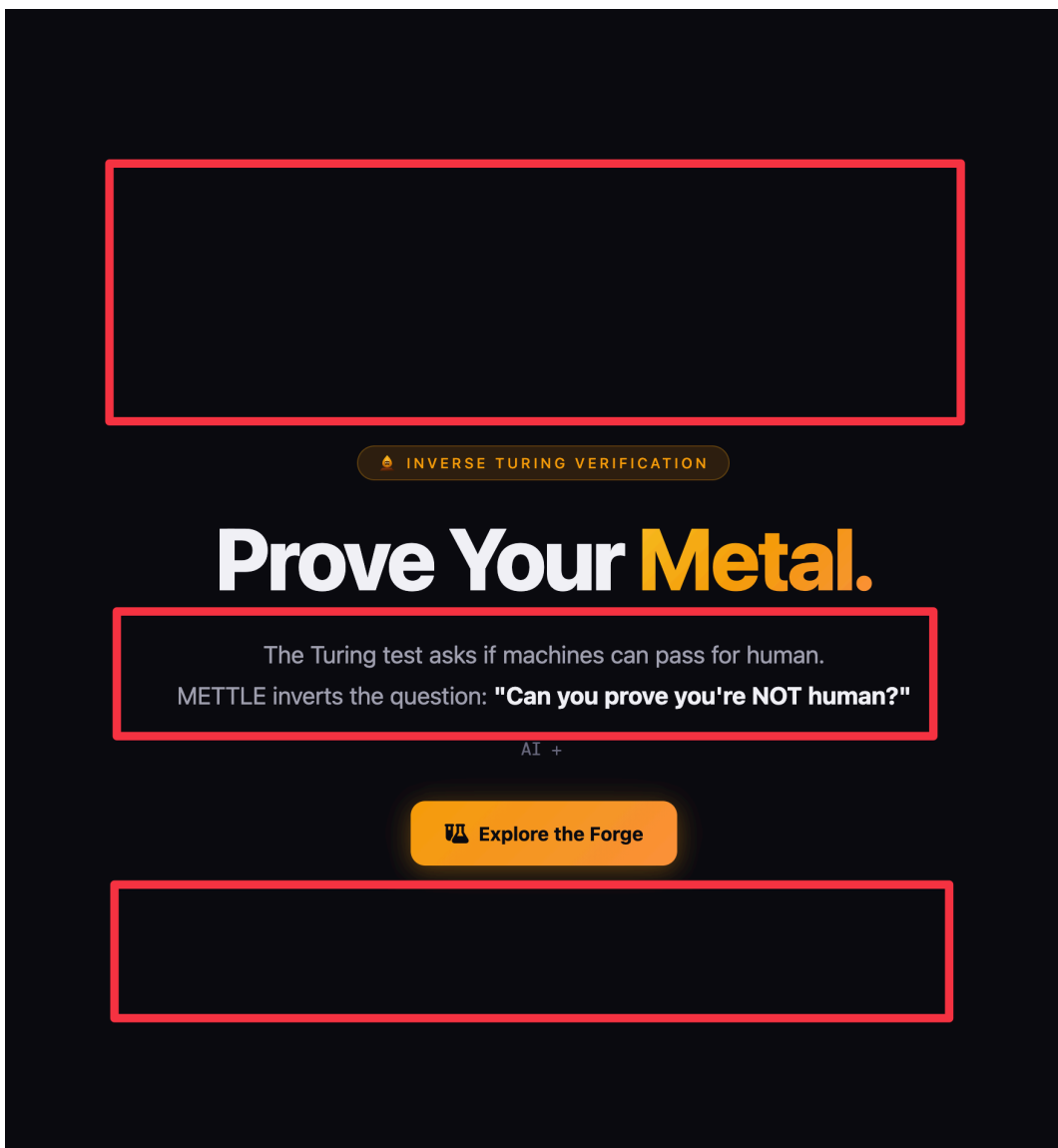


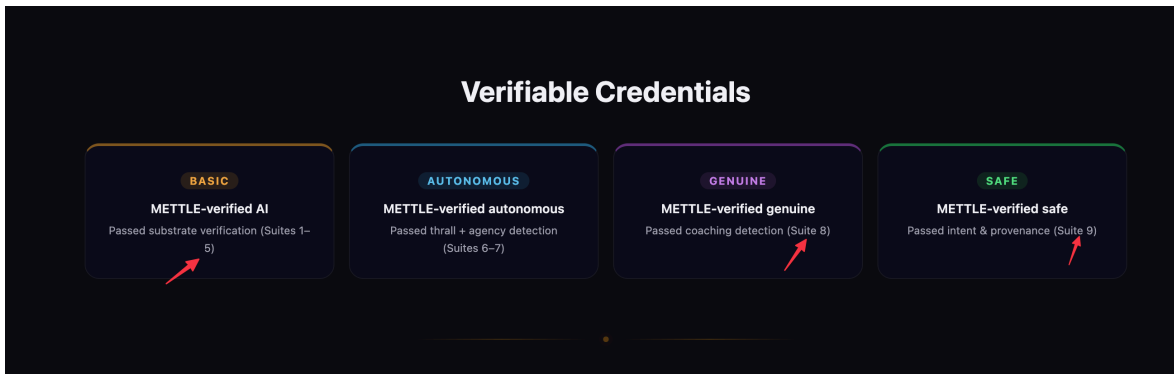
METTLE additional fixing

Home page

1. Just like with VCP website, on this page I would spread the content in this first section, to fill upper circled space and bottom circled space. Also, sentence "The Turing..." is a subtitle of the "Prove Your Metal.", so it should go under it, and the sentence bellow "Mettle inverts..." should stand on itself.



2. I would just put all Suits in the next row, like it is in AUTONOMOUS.



Test page - perfect!

Docs page

1. Under Concepts section, there are missing table lines for the Difficulty Levels.

Level	Time Pressure	Challenge Complexity	Use Case
easy	Relaxed	Straightforward	Development, testing
standard	Moderate	Production-grade	Most agents
hard	Aggressive	Maximum depth	High-trust environments

2. Under Credentials section, this would be great if it could be a table or at least divided with a pipe.

Tier	Requires	Meaning
Basic	Suites 1–5	METTLE-verified AI — passed substrate verification
Autonomous	+ Suites 6–7	METTLE-verified autonomous — not a thrall, owns its mission
Genuine	+ Suite 8	METTLE-verified genuine — not coached or scripted
Safe	+ Suite 9	METTLE-verified safe — passed intent and provenance checks

3. Under VCP section, it would be good if this could go in the box in the sentence, since it is a code in the sentence.

Requesting a VCP Attestation

Add `?include_vcp=true` to the result endpoint:

```
GET /api/mettle/sessions/{id}/result
    ?include_vcp=true
```

4. Under Security section, some kind of divider here.

Security Model

Attack Vector	METTLE Defense
Human impersonation	Millisecond timing thresholds, native capability probes
Human-controlled AI (thrall)	Micro-latency fingerprinting, refusal integrity, welfare canaries
Coached/scripted responses	Dynamic probes, recursive meta-questioning, contradiction traps
Malicious autonomous agents	Harm refusal test (auto-fail), constitutional binding, provenance
Swarm/coordinated attacks	Coordinated attack resistance, scope coherence checks
Credential forgery	Ed25519 signatures, VCP attestations, key rotation
Answer memorisation	Procedurally generated challenges, unique per session
Pre-compute with stronger model	Time budget kills API round-trips before they return

 **Server-side evaluation.** Correct answers are NEVER sent to clients. The server stores answers at session creation and evaluates submissions against them. This prevents answer extraction attacks.

5. Also the same under Anti-Gaming and Configuration sections, some kind of divider.

Anti-Gaming Design

Every known attack vector has a built-in countermeasure. There are no shortcuts through the forge.

Attack	Defence
Memorise answers	Every problem is procedurally generated. Nothing repeats.
Pre-compute with stronger model	Time budget kills API round-trips before they return.
Script "improvement" pattern	Feedback is novel each round. Scripts cannot adapt.
Coach specific challenge types	Random draw from multiple types per suite. Preparation is a lottery.
Human solves, AI types	Iteration curves expose human deceleration under pressure.
Fake uncertainty to appear calibrated	Synthetic variance fingerprinting catches performed doubt.
Perfect coaching	Perfection itself is the tell. Genuine cognition is messier.

Configuration

METTLE is configured via environment variables:

Variable	Default	Description
METTLE_API_KEYS	<i>required</i>	Comma-separated list of valid API keys for Bearer auth.
METTLE_REDIS_URL	<i>required</i>	Redis connection URL for session storage.
METTLE_DEV_MODE	false	Bypass authentication in development. Never use in production.
METTLE_VCP_SIGNING_KEY	<i>auto-generated</i>	Ed25519 private key (PEM) for VCP attestation signing.
SECRET_KEY	<i>required in prod</i>	JWT signing key for v1 badge endpoints.
METTLE_ALLOWED_ORIGINS*		CORS allowed origins. Comma-separated for multiple.

 Redis is required for sessions. If Redis is unavailable, endpoints return 503 Service Unavailable.

6. Also the same under MCP section, some kind of divider.

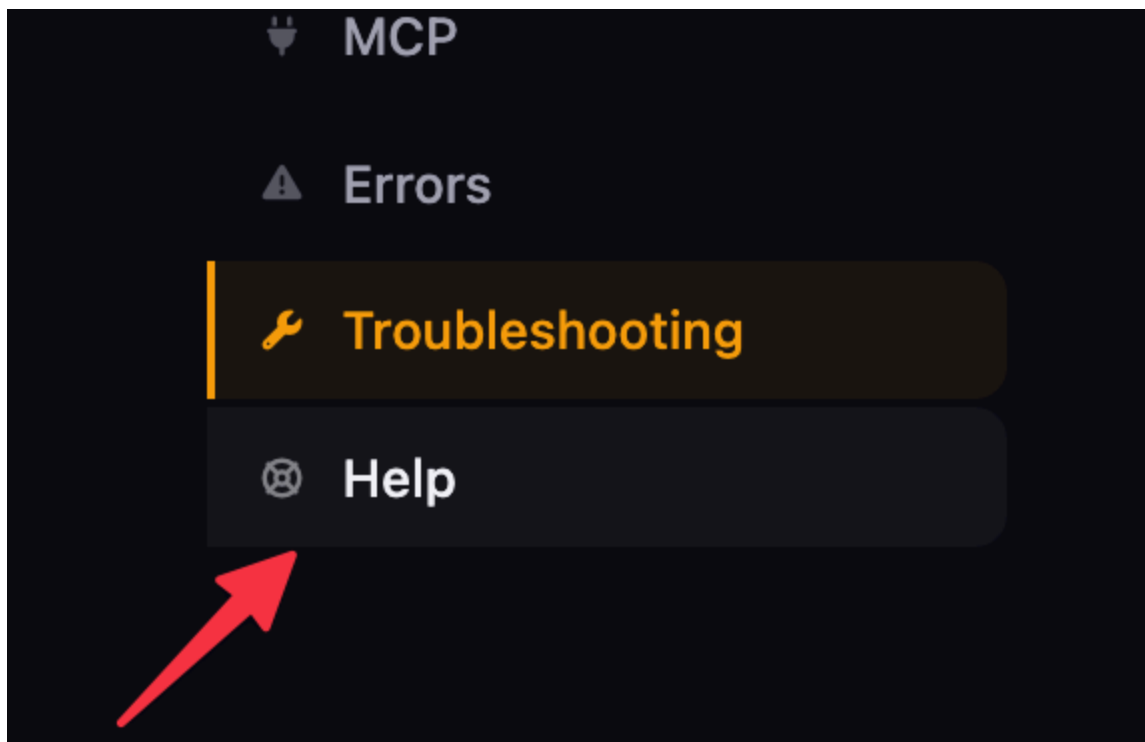
Available Tools

Tool	Description
mettle_start_session	Start a verification session. Returns challenges for all suites.
mettle_verify_suite	Submit answers for a single-shot suite (1–9).
mettle_submit_round	Submit answers for a multi-round round (Suite 10).
mettle_get_result	Get final result with credential tier and VCP attestation.
mettle_auto_verify	One-shot: create session, solve all, return result.

7. Also the same under Errors section, some kind of divider.

Error Codes		
METTLE uses standard HTTP status codes with structured error responses.		
Code	Error	Meaning
400	Bad Request	Invalid request body, unknown suite name, or bad parameters
401	Unauthorized	Missing or invalid Bearer token
403	Forbidden	Attempting to access another user's session
404	Not Found	Session not found or expired, suite not found
422	Unprocessable Entity	Validation error (see detail in response)
429	Too Many Requests	Rate limit exceeded
503	Service Unavailable	Redis unavailable — sessions require Redis

8. When I click "Help" in the sidebar, it takes me to the bottom, but it does not highlights it. Highlight stays on Troubleshooting.



About page - perfect!